

# AI for Climate, and Generative AI for climate downscaling (Hackathon)

Advanced School in Applied Machine Learning:  
Uncertainty and Robustness of DL Models for  
Science



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for Theoretical Physics**



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International Atomic Energy Agency



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United Nations  
Educational, Scientific  
and Cultural Organization

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Instituto de Física de Cantabria



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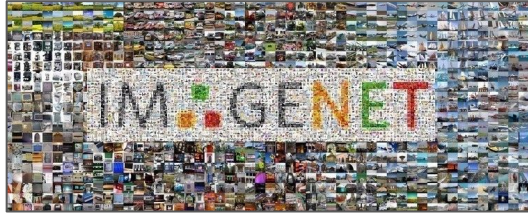
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# Benchmark

A benchmark in DL is a **standardized dataset** and **evaluation metric** used to objectively compare and **track the performance of different models**

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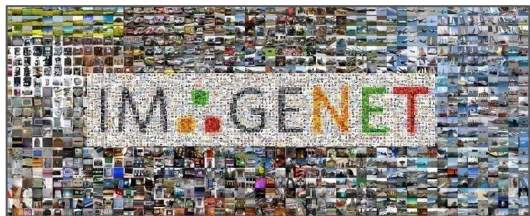
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There are multiple benchmarks for **LLMs** that try to **evaluate different dimensions**, such as coding, reasoning, or mathematical skills

## LiveBench

### A Challenging, Contamination-Free LLM Benchmark

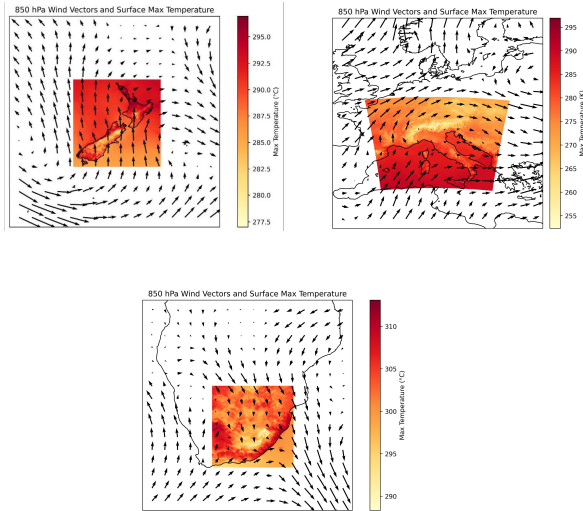
LiveBench will appear as a [Spotlight Paper](#) in ICLR 2025.

This work is sponsored by [Abacus.AI](#)



| Model                                    | Organization | Global Average | Reasoning Average | Coding Average | Agentic Coding Average |
|------------------------------------------|--------------|----------------|-------------------|----------------|------------------------|
| <a href="#">GPT-5 High</a>               | OpenAI       | 78.59          | 98.17             | 75.31          | 43.33                  |
| <a href="#">GPT-5 Medium</a>             | OpenAI       | 76.45          | 96.58             | 73.25          | 35.00                  |
| <a href="#">GPT-5 Low</a>                | OpenAI       | 75.34          | 90.47             | 72.49          | 41.67                  |
| <a href="#">o3 Pro High</a>              | OpenAI       | 74.72          | 94.67             | 76.78          | 31.67                  |
| <a href="#">o3 High</a>                  | OpenAI       | 74.61          | 94.67             | 76.71          | 36.67                  |
| <a href="#">Claude 4.1 Opus Thinking</a> | Anthropic    | 73.48          | 93.19             | 73.96          | 33.33                  |

**CORDEX ML-Bench** is a **benchmark designed to evaluate the performance of machine learning-based climate downscaling** models across different regions covering both the standard (perfect prognosis ESD) and emulation climate downscaling approaches



Different regions

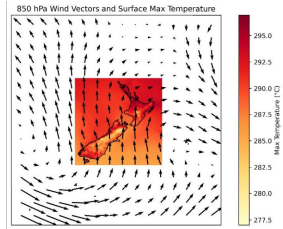
Different domains

Different training experiments

Different evaluation experiments



## New Zealand

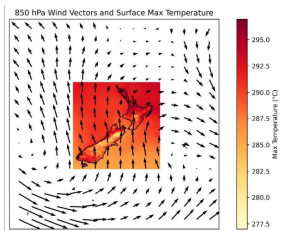


The objective of the hackathon is to train a **deep learning** model for the statistical **downscaling of precipitation (RCM emulation)**

**Training Period:** 1961-1980 + 2080-2099

**Test Period:** 2041-2060

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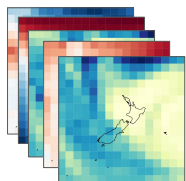
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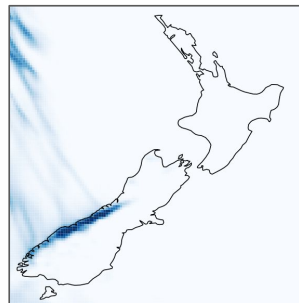
### Predictors



- Meridional Wind Comp.
- Zonal Wind Comp.
- Specific Humidity
- Geopotential Height
- Air Temperature

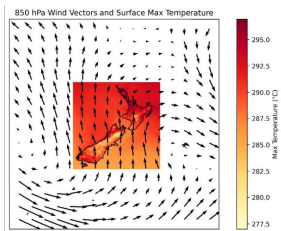
850, 700 and 500 hPa

### Predictand



Precipitation

## New Zealand



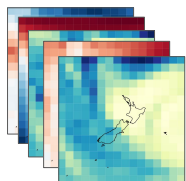
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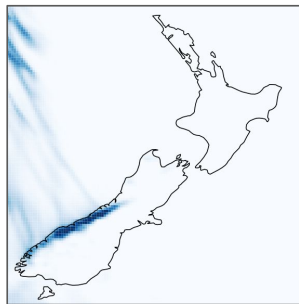
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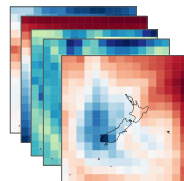
**Predictand**



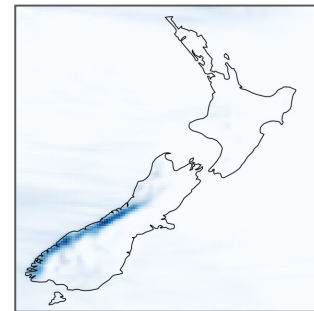
Precipitation

**Test** (2041-2060)

**Predictors**



**Predictand**





<https://github.com/jgonzalezab/ICTP-Downscaling-Hackathon-student>



README

## ICTP Downscaling Hackathon — Student Kit

Train a deep learning model for statistical downscaling of precipitation over New Zealand, then submit predictions for the **Perfect Interpolation** experiment (2041–2060).

Evaluation and the leaderboard are run by the organizers. Your deliverable is a single NetCDF file.

### Exercise summary

| Item                    | Value                                        |
|-------------------------|----------------------------------------------|
| Domain                  | New Zealand (NZ)                             |
| Variable                | Precipitation ( pr , mm/day)                 |
| Training experiment     | RCM Emulator ( Emulator_hist_future )        |
| Training period         | 1961–1980 + 2080–2099                        |
| Evaluation              | Perfect Interpolation, mid-century 2041–2060 |
| Predictors at test time | Perfect (coarsened RCM), ACCESS-CM2          |

Your goal: learn the mapping from coarse predictors (16×16, 15 atmospheric variables) to high-resolution daily precipitation (128×128).

### 1. Data

Student data on Leonardo:

```
/leonardo/home/userexternal/jgonz14/NZ_domain_student
```

Layout:

```
NZ_domain_student/  
├─ train/Emulator_hist_future/  
│ └─ predictors/ACCESS-CM2_1961-1980_2080-2099.nc  
│ └─ predictors/Static_fields.nc # optional orography  
├─ target/pr_tasmax ACCESS-CM2_1961-1980_2080-2099.nc  
└─ test/mid_century/predictors/perfect/ACCESS-CM2_2041-2060.nc
```

You do not receive the mid-century ground-truth target, that is used only for evaluation.

### Quick inspection with xarray

```
import xarray as xr  
  
data_root = "/leonardo/home/userexternal/jgonz14/NZ_domain_student"  
predictors = xr.open_dataset(f"{data_root}/train/Emulator_hist_future/predictors/ACCESS-CM2_1961-1980-2080-2099.nc")  
target = xr.open_dataset(f"{data_root}/train/Emulator_hist_future/target/pr_tasmax_ACCESS-CM2_1961-1980-2080-2099.nc")  
print(predictors.dims) # time=14000, lat=16, lon=16  
print(target["pr"].dims) # time=14000, lat=128, lon=128
```

### 2. Install dependencies (Leonardo)

These steps assume you already have a Conda init script at `$HOME/Conda_init.txt`. If you do not, copy `testing/`